QSAO Case Competition: Arizona Diamondbacks

Competitive Roster Construction

By: Arvin A, Jordan B, Rohan B, Rocco D

Team Overview

Last Three Seasons:
2025: 80-82, 4th NL West & Missed Playoffs
2024: 89-73, 3rd NL West & Missed Playoffs
2023: 84-78, 2nd NL West & Lost World Series

Current Situation – Contending:
- Entering a clear contending window (‘26-’28) driven by an offensive core with some players entering their prime and some about to age out
- Team has a consistently strong offence that can hit for contact and power
- Above-average speed and defence
- Pitching is a key area of concern, especially the thin and inexperienced bullpen, and must be addressed via trade in order for the team to be successful
- Could use a depth upgrade at Starter and First Base, with additional bats to help improve the DH spot and allow for positional flexibility

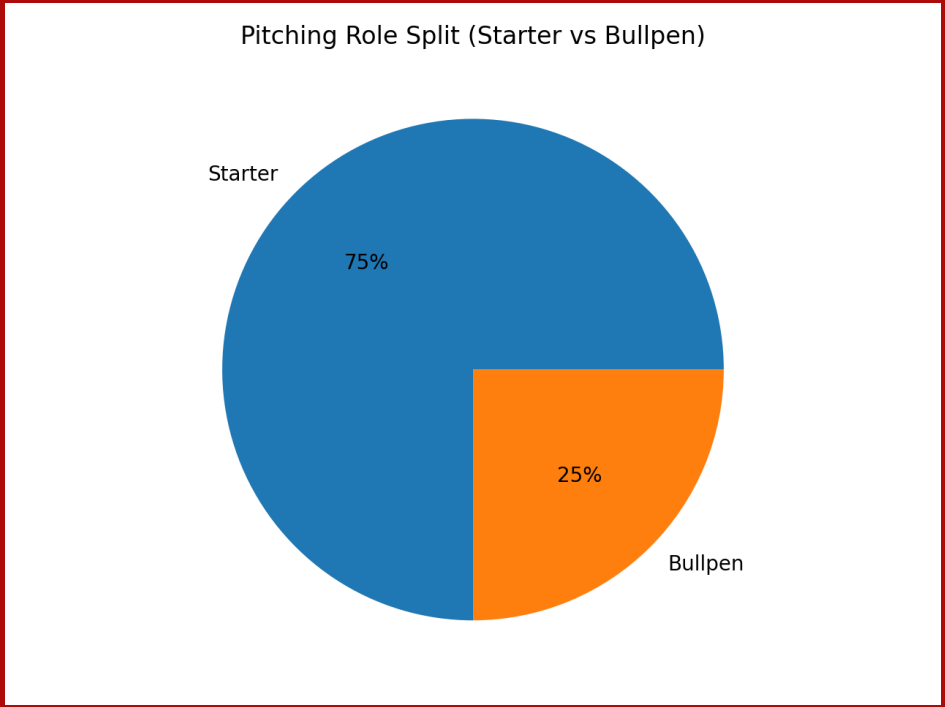
Key Players:
- Geraldo Perdomo, SS, 7.0 WAR, Age 26
- Corbin Carroll, RF, 5.8 WAR, Age 25
- Ketel Marte 2B, 4.4 WAR, Age 32

2026 Payroll:
$171 Million (14th of 30)

Positional Strengths:
- Middle Infield (2B/SS)
- Catcher
- Right Field

Positional Weaknesses:
- Bullpen
- First Base/DH
- Depth Starter

Roster and Performance Challenges



The Diamondbacks' primary challenges are bullpen reliability, lineup depth beyond the core, and converting replaceable surplus into high-leverage value while protecting the contending window.

Strategic Objectives Moving Forward

Future Transactions:
- Opt-out of Gurriel Jr. & Decline Mutual Option for Soroka in 2027
- Consider re-signing the veteran McCann at catcher, rest can walk
- Rest of core is signed long-term

Key Objectives:
- Maximize the Contending Window
- Upgrade Bullpen via Targeted Trades
- Protect the Core, Trade/Upgrade the Margins
- Optimize Roles, Versatility, and Balance

Players With Guaranteed Salaries ▾

Player	Age	Service Time	Contract	AAV	2026	2027	2028	2029	2030	2031	2032
Corbin Burnes	31.3	7.049	6 yr, $210M (2025-30); can opt out after 2026	$33,041,322	$35,000,000	$35,000,000	$35,000,000	$35,000,000	$35,000,000	FREE AGENT	
Nolan Arenado	34.8	12.155	9 yr, $275M (2019-27)	$21,000,000	$27,000,000	$15,000,000	FREE AGENT				
Eduardo Rodriguez	32.8	10.070	4 yr, $80M (2024-27), 2028 mutual option; converts to vesting if IP conditions are met	$20,000,000	$21,000,000	$19,000,000	$17,000,000				
Merrill Kelly	37.3	7.000	2 yr, $40M (2026-27), 2028 vesting option	$20,000,000	$18,000,000	$22,000,000	$12,000,000				
Ketel Marte	32.3	9.162	7 yr, $116.5M (2025-30), 2031 player option; deferrals lower AAV	$14,640,966	$15,000,000	$12,000,000	$20,000,000	$22,000,000	$22,000,000	$11,500,000	
Lourdes Gurriel Jr.	32.3	7.094	3 yr, $42M (2024-26), 2027 club option	$14,000,000	$13,000,000	$14,000,000					
Corbin Carroll	25.4	3.038	8 yr, $111M (2023-30), 2031 club option	$13,875,000	$10,625,000	$12,625,000	$14,625,000	$28,625,000	$28,625,000	$28,000,000	
Geraldo Perdomo	26.3	4.015	4 yr, $45M (2026-29), 2030 club option	$11,250,000	$6,250,000	$9,250,000	$12,250,000	$14,250,000	$15,000,000		
Brandon Pfaadt	27.3	2.097	5 yr, $45M (2026-30), 2031 club option, 2032 mutual option	$9,000,000	$3,400,000	$5,400,000	$8,400,000	$11,400,000	$15,400,000	$21,000,000	$25,000,000
Michael Soroka	28.5	7.009	1 yr, $7.5M (2026), 2027 mutual option	$7,500,000	$6,500,000	$10,000,000					
Justin Martinez	24.5	2.024	5 yr, $18M (2025-29), 2030-31 club options	$3,600,000	$2,400,000	$3,400,000	$4,400,000	$5,900,000	$7,000,000	$9,000,000	
James McCann	35.6	10.126	1 yr, $2.75M (2026)	$2,750,000	$2,750,000	FREE AGENT					
Taylor Clarke	32.7	5.148	1 yr, $1.55M (2026)	$1,550,000	$1,550,000	FREE AGENT					

Salaries for players heading to an arbitration hearing are shown as the midpoint of the figures the team and player filed.

Players Eligible For Arbitration ▾

Player	Age	Service Time	Contract	AAV	2026	2027	2028	2029	2030	2031	2032
Ryan Thompson	33.6	5.095	1 yr, $3.95M (2026)		$3,950,000	FREE AGENT					
A.J. Puk	30.8	5.124	1 yr, $3.1M (2026)		$3,100,000	FREE AGENT					
Kevin Ginkel	31.9	5.033	1 yr, $2.725M (2026)		$2,725,000	FREE AGENT					
Pavin Smith	30.0	4.015	1 yr, $2.25M (2026)		$2,250,000	ARB 3	FREE AGENT				
Ryne Nelson	28.0	3.020	1 yr, $3M (2026)		$3,000,000	ARB 2	ARB 3	FREE AGENT			
Gabriel Moreno	26.0	3.061	1 yr, $2.55M (2026)		$2,550,000	ARB 2	ARB 3	FREE AGENT			
Alek Thomas	25.8	3.103	1 yr, $1.963M (2026)		$1,962,500	ARB 2	ARB 3	FREE AGENT			

Data Processing And Analysis

BR Baseball Data

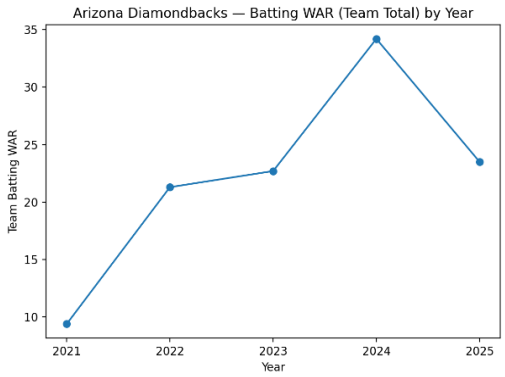
- WAR, playing time, positions, age, plate appearances/innings pitched, and contract status

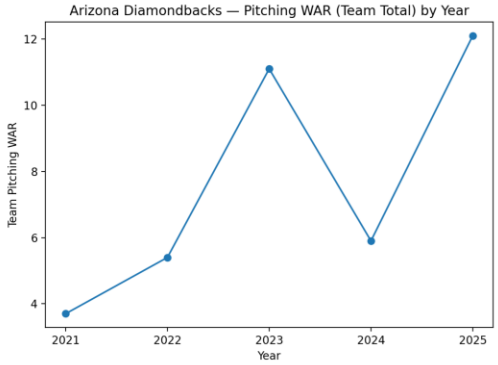
Statcast Metrics

- xwOBA, hard-hit %, barrel rate, and pitch characteristics

Allowed us to evaluate player value, role, sustainability and underlying skills independent of results.

Separates true talent from short-term variance, allowing us to obtain a more accurate overview of each player.

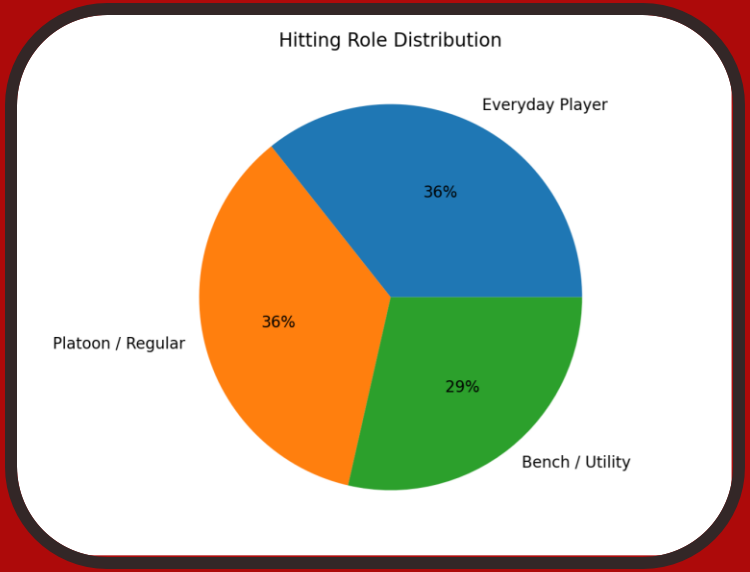




Data Processing And Analysis

Data Filtering

Step 1
Filtered league-wide datasets to Diamondback players only



Step 2
Organized players and assigned roles (starter, reliever, everyday player)

Core Player
Player WAR >= WAR Avg
Role = "Everyday Player"
Age <= 32

Surplus Player
Player WAR < WAR Avg
Role = "Bench / Utility"
Age >= 27

Step 3
Incorporated a player age factor to help distinguish between core and surplus players

Player-Evaluation Methodology & Data Pipeline

1) Data Integration
   - Integrated **Baseball-Reference** and **Statcast** data (2021–2025)
2) Data Cleaning & Standardization
   - Filtered and standardized team-level and player-level datasets
3) Feature Engineering
   - Positional and defensive adjustments applied to reflect roster scarcity
   - Role context added (starter vs bullpen, everyday vs platoon)

4) Projection & Role Classification
   - WAR based machine learning model (regularized regression) used to project 2025 WAR
   - Python using Pandas and NumPy for data processing, Matplotlib for Data Visualization, and Scikit-Learn for Machine Learning
   - Players classified into functional archetypes and roster roles

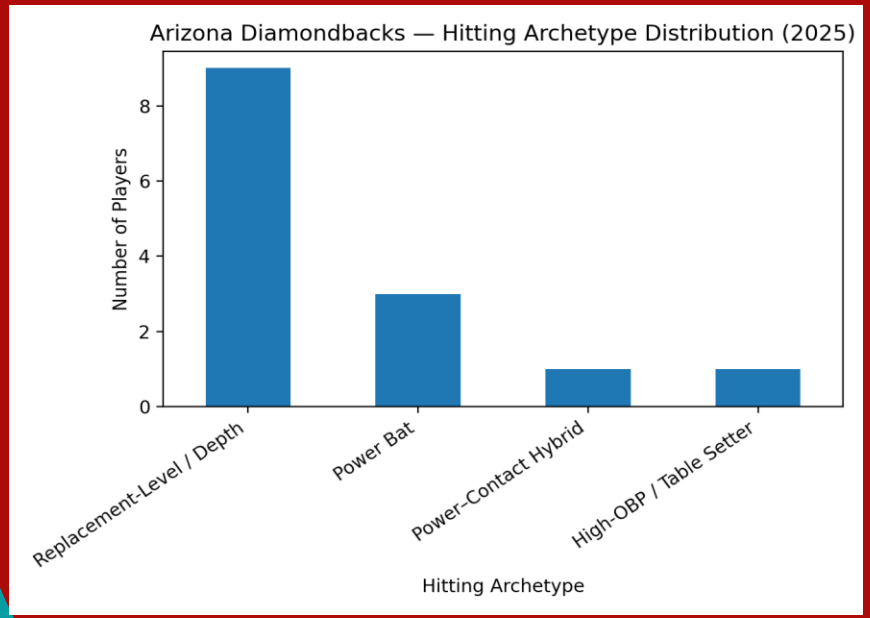
Player-Evaluation Framework: Metrics & Archetype

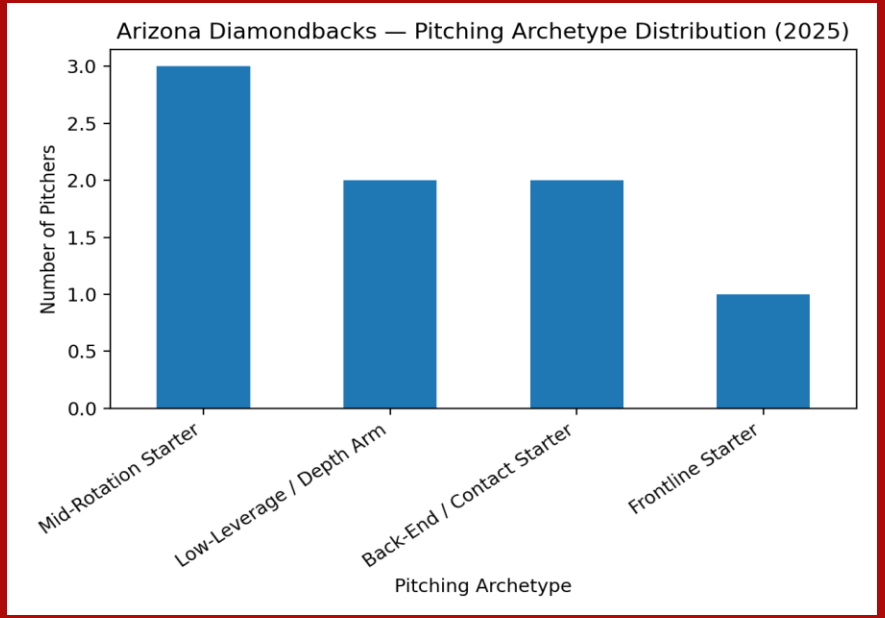
Quantitative Metrics:

- **WAR** as a unified measure of total on-field value
- Offense: OBP, SLG, ISO, contact-quality indicators
- **Pitching:** ERA-style run prevention, strikeout and contact suppression
- Usage & workload indicators (starter vs bullpen, leverage)

Target formulas**:**

- Batters: WAR_per_600 = WAR / PA * 600
- Pitchers: WAR_per_180 = WAR / IP * 180





Using the Model to Drive Roster Decisions

Trades:

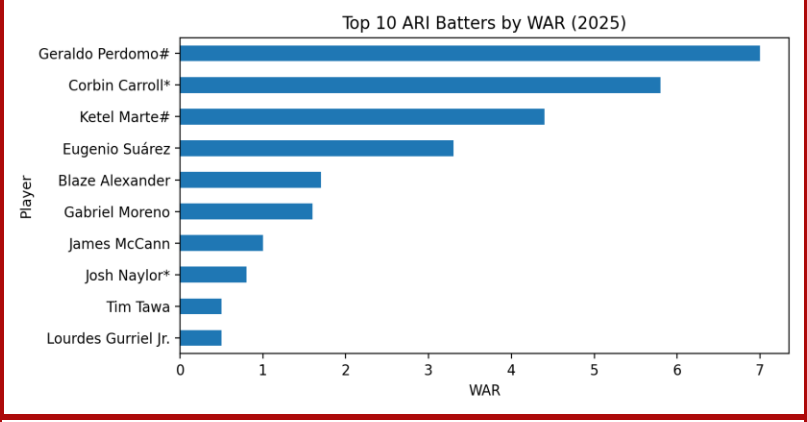
- Identified surplus players by archetype and role
- Targeted specific skill gaps (e.g., bullpen swing-and-miss) rather than star names
- Optimized for marginal WAR gain per acquisition cost

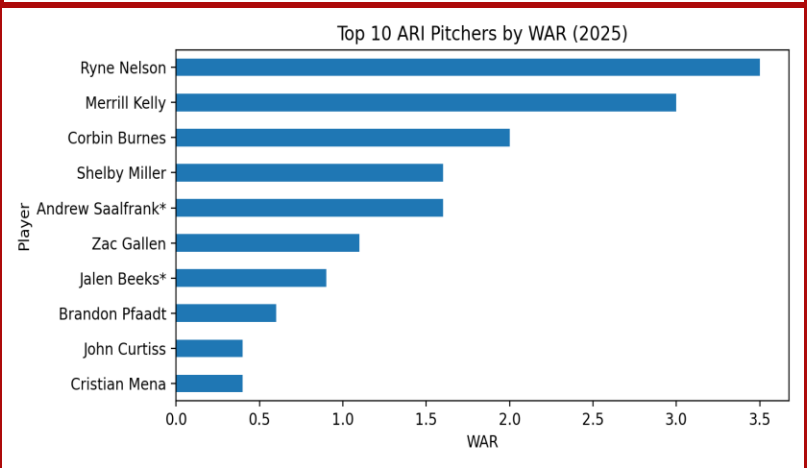
Extensions

- Prioritized prime-age players with stable projected WAR
- Minimized long-term risk by limiting commitments to high-variance roles

Player Development:

- Retained prospects in scarce roles (SP, leverage relief)
- Allocated playing time by role fit, not reputation





Trade Proposal #1 – The Bullpen

with San Diego Padres

IN:

Jason Adam

Closer
2.4 WAR,
1.93 ERA
In 2025

David Morgan

Reliever
1.2 WAR,
2.66 ERA
In 2025

OUT:

Taylor Clarke

Reliever
1.0 WAR,
3.25 ERA
In 2025

Brandyn Garcia

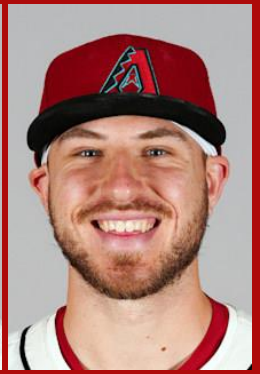
Reliever
-0.2 WAR,
5.65 ERA
In 2025

LuJames Groover

Infielder
2nd Round Pick
In 2023

Yilber Diaz

Pitcher
0.2 WAR,
4.31 ERA
In Career



AJ Vukovich

INF/OF
4th Round Pick
In 2020

Trade Proposal #2 – 1B + SP Depth

with Baltimore Orioles

IN:

Ryan Mountcastle

1B/LF/DH
2.6 WAR,
.733 OPS
In 2024

Dietrich Enns

Starting Pitcher
0.4 WAR,
4.08 ERA
In 2025

OUT:

Adrian Del Castillo

1B/C/DH
-0.1 WAR,
.682 OPS
In 2025

Tim Tawa

OF/2B/1B
0.5 WAR,
.620 OPS
In 2025

Mitch Bratt

Starting Pitcher
5th Round Pick
In 2021

Eduardo Rodriguez

Starting Pitcher
0.3 WAR,
5.02 ERA
In 2025

+

$15 Million To Cover $40 Million Owed Across 2026-2027 For Eduardo Rodriguez

Updated Roster and Payroll Figure

Lineup:

1. SS Geraldo Perdomo (S)
2. 2B Ketel Marte (S)
3. RF Corbin Carroll (L)
4. DH Lourdes Gurriel Jr. (R) – **IR**
5. C Gabriel Moreno (R)
6. 3B Nolan Arenado (R)
7. 1B Ryan Mountcastle (R) – **New Addition**
8. CF Alek Thomas (L)
9. LF Blaze Alexander (R)

Bench Bats:

1B/OF Pavin Smith (L)
C James McCann (R)
INF Jordan Lawlar (R)
OF Ryan Waldschmidt (R) – **Minors**

Rotation:

1. Corbin Burnes (R)
2. Merrill Kelly (R)
3. Ryne Nelson (R)
4. Dietrich Enns (L) – **New Addition**
5. Brandon Pfaadt (R)

Bullpen:

Jason Adam (R) – **New Addition**
David Morgan (R) – **New Addition**
Andrew Saalfrank (L)
Ryan Thompson (R)
Kevin Ginkel (R)
Jonathan Loasiga (R)
Drey Jameson (R)
A.J. Puk (L)

Depth Options:

INF Tommy Troy (R) – **Minors**
SP Michael Soroka (R)
SP/RP Cristian Mena (R) - **Minors**

ESTIMATED 2026 TOTAL PAYROLL:

$157 Million
(reduction of about $14 M by bringing in Mountcastle and Adam while trading Rodriguez and cash; under $244M luxury tax, 15th of 30)

Future Steps

Continued Roster Optimization:

- Continuously re-evaluate roster using updated WAR projections and archetype balance
- Monitor bullpen leverage performance and optimize usage based on run expectancy
- Identify emerging surplus positions early to maintain trade flexibility
- Reassess lineup sequencing to maximize OBP and run creation efficiency

Long-Term Development Strategy:

- Prioritize development of starting pitching and high-leverage relief, where external acquisition cost is highest
- Maintain pipeline of versatile hitters to reduce WAR concentration risk on solely being power or contact-focused
- Integrate injury-risk and workload management models into player evaluations
- Align prospect development timelines and willingness to trade young players with the team's competitive window from '26-'28

Thank You!